① a) Fun. de verosimilitud $L(\theta)$

Para construir $\mathcal{L}(\theta)$ necesitamos conocer $p(y_i | x_i; \theta)$ (el cual está dado en el problema como una distribución de Poisson). Aplicamos la def. de $\mathcal{L}(\theta)$

$$\mathcal{L}(\theta) = \prod_{i=1}^{m} p_i(y_i | x_i ; \theta)$$

$$\mathcal{L}(\theta) = \prod_{i=1}^{m} \frac{h_\theta(x_i)^{y_i} e^{-h_\theta(x_i)}}{y_i!} \quad \text{con } h_\theta(x_i) = \theta_0 + \theta_1 x_i$$

$$\mathcal{L}(\theta) = \prod_{i=1}^{m} \frac{(\theta_0 + \theta_1 x_i)^{y_i} - e^{-(\theta_0 + \theta_1 x_i)}}{y_i!}$$

b) log-verosimilitud $\ell(\theta)$ y fun. de coste

Aplicamos $\ln \mathcal{L}(\theta)$:

$$\ln \mathcal{L}(\theta) = \ln \left[\prod_{i=1}^{m} \frac{h_\theta(x_i)^{y_i} e^{-h_\theta(x_i)}}{y_i!} \right]$$

$$\ln \mathcal{L}(\theta) = \sum_{i=1}^{m} \left\{ y_i \ln [h_\theta(x_i)] - h_\theta(x_i) - \ln(y_i!) \right\}$$

llamamos $\ell(\theta) = \ln \mathcal{L}(\theta)$ y usamos $h_\theta(x_i) = \theta_0 + \theta_1 x_i$

$$\ell(\theta) = \sum_{i=1}^{m} \left\{ y_i \ln(\theta_0 + \theta_1 x_i) - [\theta_0 + \theta_1 x_i] - \ln(y_i!) \right\}$$

Para la fun. de coste: $J(\theta) = -\ell(\theta)$

$$J(\theta) = \sum_{i=1}^{m} \left\{ \theta_0 + \theta_1 x_i + \underbrace{\ln(y_i!)}_{\text{como } J(\theta) \text{ solo depende de parámetros } \theta. \text{ despreciamos } \ln(y_i!)} - y_i \ln(\theta_0 + \theta_1 x_i) \right\}$$

c)

De esta manera, llegamos a que la fun. de coste es:

$$J(\theta) = \sum_{i=1}^{m} \left\{ \theta_0 + \theta_1 x_i - y_i \ln(\theta_0 + \theta_1 x_i) \right\}$$

$$J(\theta) = \sum_{i=1}^{m} \left\{ h_\theta(x_i) - y \ln [h_\theta(x_i)] \right\}$$

d) Calcular el gradiente de $J(\theta)$ respecto a θ_0 y θ_1.

Como $J(\theta) = \sum_{i=1}^{m} [\theta_0 + \theta_1 x_i - y_i \ln [\theta_0 + \theta_1 x_i]]$

calculamos $\partial J / \partial \theta_1$ y $\partial J / \partial \theta_0$

$$\rightarrow \frac{\partial J}{\partial \theta_0} = \frac{\partial}{\partial \theta_0} \left\{ \sum_{i=1}^{m} \theta_0 + \theta_1 x_i - y_i \ln [\theta_0 + \theta_1 x_i] \right\}$$

$$= \sum_{i=1}^{m} \left\{ \frac{\partial}{\partial \theta_0} \theta_0 - y_i \frac{\partial}{\partial \theta_0} \ln [\theta_0 + \theta_1 x_i] \right\}$$

$$= \sum_{i=1}^{m} \left\{ 1 - y_i \cdot \frac{1}{\theta_0 + \theta_1 x_i} (1) \right\}$$

$$= \sum_{i=1}^{m} \left\{ 1 - \frac{y_i}{h_\theta(x_i)} \right\} \longrightarrow \theta_0^{K+1} = \theta_0^K - \alpha \sum_{i=1} \left[1 - \frac{y_i}{\theta_0^K - \theta_1^K x_i} \right]$$

(tasa aprendizaje)

$$\rightarrow \frac{\partial J}{\partial \theta_1} = \sum_{i=}^{m} \left\{ \frac{\partial}{\partial \theta_1} (\theta_1 x_i) - y_i \frac{\partial}{\partial \theta_1} \ln [\theta_0 + \theta_1 x_i] \right\}$$

$$= \sum_{i=1}^{m} \left\{ x_i - y_i \frac{1}{\theta_0 + \theta_1 x_i} (x_i) \right\}$$

$$= \sum_{i=1}^{m} \left\{ x_i - \frac{y_i x_i}{h_\theta(x_i)} \right\}$$

$$= \sum_{i=1}^{m} x_i \left\{ 1 - \frac{y_i}{h_\theta(x_i)} \right\} \longrightarrow \theta_1^{K+1} = \theta_1^K - \alpha \sum_{i=1}^{n} x_i \left[1 - \frac{y_i}{\theta_0^K - \theta_1^K x_i} \right]$$

e) Interpretar $h_\theta(x)$, θ_0 y θ_1 en el contexto del detector.

- $h_\theta(x)$: número esperado de partículas detectadas para una condición experimental x.
- θ_0: tasa esperado de detección cuando $x = 0$
- θ_1: cambio del número de detecciones cuando se modifica x.

$\sum_{i=1}$

(2)

a) Calcular el error de la capa de salida $\delta_{i,1}^{[2]} = \frac{\partial J_i}{\partial z_{i,1}^{[2]}}$

$$\frac{\partial J_i}{\partial z_{i,1}^{[2]}} = \frac{\partial}{\partial z_{i,1}^{[2]}} \left[e^{z_{i,1}^{[2]}} - y_i z_{i,1}^{[2]} \right]$$

$$\frac{\partial J_i}{\partial z_{i,1}^{[2]}} = e^{z_{i,1}^{[2]}} - y_i \quad \text{como} \quad h_\theta(x_i) = e^{z_{i,1}^{[2]}} \Rightarrow \delta_{i,1}^{[2]} = h_\theta(x_i) - y_i$$

b) Calcular los grad. respecto a los parámetros de la capa de salida:

$$\frac{\partial J_i}{\partial \theta_{1,1}^{[2]}} \quad \text{y} \quad \frac{\partial J_i}{\partial b_1^{[2]}}$$

$$\rightarrow \frac{\partial J_i}{\partial \theta_{1,1}^{[2]}} = \frac{\partial J_i}{\partial z_{i,1}^{[2]}} \frac{\partial z_{i,1}^{[2]}}{\partial \theta_{1,1}^{[2]}}$$

✓ de a) sabemos $\frac{\partial J_i}{\partial z_{i,1}^{[2]}} = \delta_{i,1}^{[2]}$

✓ $\frac{\partial z_{i,1}^{[2]}}{\partial \theta_{1,1}^{[2]}} = \frac{\partial}{\partial \theta_{1,1}^{[2]}} \left[\theta_{1,1}^{[2]} A_{i,1}^{[1]} + b_1^{[2]} \right]$

$\frac{\partial z_{i,1}^{[2]}}{\partial \theta_{1,1}^{[2]}} = A_{i,1}^{[1]}$

Retomamos

$$\frac{\partial J_i}{\partial \theta_{1,1}^{[2]}} = \delta_{i,1}^{[2]} A_{i,1}^{[1]}$$

$$= [h_\theta(x_i) - y_i] A_{i,1}^{[1]}$$

$$\rightarrow \frac{\partial J_i}{\partial b_1^{[2]}} = \frac{\partial J_i}{\partial z_{i,1}^{[2]}} \frac{\partial z_{i,1}^{[2]}}{\partial b_1^{[2]}}$$

$\delta_{i,1}^{[2]}$

$$\rightarrow \frac{\partial}{\partial b_1^{[2]}} \left(\theta_{11}^{[2]} A_{i,1}^{[1]} + b_1^{[2]} \right] = 1$$

$$\frac{\partial J_i}{\partial b_1^{[2]}} = \delta_{i,1}^{[2]} (1) = h_\theta(x_i) - y_i$$

c) Calcular el error que se propaga hacia la capa oculta

$$\delta_{i,1}^{[1]} = \frac{\partial J_i}{\partial z_{i,1}^{[1]}} = \frac{\partial J_i}{\partial z_{i,1}^{[2]}} \frac{\partial z_{i,1}^{[2]}}{\partial A_{i,1}^{[1]}} \frac{\partial A_{i,1}^{[1]}}{\partial z_{i,1}^{[1]}}$$

$\delta_{i,1}^{[2]}$

$$\rightarrow \frac{\partial z_{i,1}^{[2]}}{\partial A_{i,1}^{[1]}} = \frac{\partial}{\partial A_{i,1}^{[1]}} \left(\theta_{1,1}^{[2]} A_{i,1}^{[1]} + b_1^{[2]}\right)$$

$$= \theta_{1,1}^{[2]}$$

$$\rightarrow \frac{\partial A_{i,1}^{[1]}}{\partial z_{i,1}^{[1]}}$$ debemos tener en cuenta que: $A_{i,1}^{[1]} = \sigma[z_{i,1}^{[1]}]$

$$\frac{\partial A_{i,1}^{[1]}}{\partial z_{i,1}^{[1]}} = \sigma'(z_{i,1}^{[1]})$$

Así $\delta_{i,1}^{[1]} = \delta_{i,1}^{[2]} \theta_{1,1}^{[2]} \sigma'(z_{i,1}^{[1]})$

$$= [h_\theta(x_i) - y_i] \theta_{1,1}^{[2]} \sigma'(z_{i,1}^{[1]})$$

d) Calcule los gradientes respecto a los parámetros de la primera capa

$$\frac{\partial J_i}{\partial \theta_{1,1}^{[1]}}, \quad \frac{\partial J_i}{\partial b_1^{[1]}}$$

$$\rightarrow \frac{\partial J_i}{\partial \theta_{1,1}^{[1]}} = \frac{\partial J_i}{\partial z_{i,1}^{[1]}} \frac{\partial z_{i,1}^{[1]}}{\partial \theta_{1,1}^{[1]}}$$

$\delta_{i,1}^{[1]}$ $\rightarrow \frac{\partial}{\partial \theta_{1,1}^{[1]}} [\theta_{1,1}^{[1]} x_i + b_i^{[1]}] = x_i$

$$\frac{\partial J_i}{\partial \theta_{1,1}^{[1]}} = \delta_{i,1}^{[1]} x_i$$

↳ ipunto c)

$$\frac{\partial J_i}{\partial \theta_{1,1}^{[2]}} = x_i [h_\theta(x_i) - y_i] \theta_{1,1}^{[2]} \sigma'[z_{i,1}^{[1]})$$

③

Part b

$$\frac{\partial J_i}{\partial b_1^{[1]}} = \frac{\partial J_i}{\partial z_{i,1}^{[1]}} \frac{\partial z_{i,1}^{[1]}}{\partial b_1^{[1]}}$$

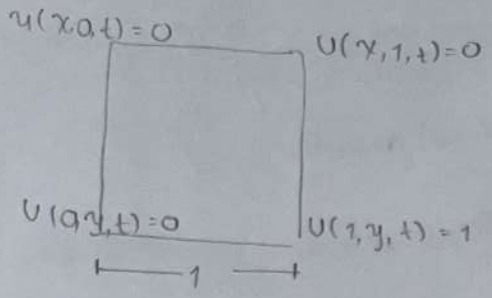
$\frac{\partial J_i}{\partial z_{i,1}^{[1]}} = \delta_{i,1}^{[1]}$

$\frac{\partial z_{i,1}^{[1]}}{\partial b_1^{[1]}} \rightarrow \frac{\partial}{\partial b_1^{[1]}} \left(\theta_{1,1}^{[1]} x_i + b_1^{[1]} \right) = 1$

$$= \delta_{i,1}^{[1]} (1)$$

$$\frac{\partial J_i}{\partial b_1^{[1]}} = \left[h_\theta(x_i) - y_i \right) \theta_{i,1}^{[2]} \sigma'(z_{i,1}^{[1]})$$

Parte analítica



cond. iniciales

(1) → $u(x,y,0) = \text{sen}(\pi x)\text{sen}(\pi y)$ (pos)

(2) → $u_t(x,y,0) = \sqrt{2}\,\pi\,\text{sen}(\pi x)\text{sen}(\pi y)$ (vel)

Sea $\frac{\partial^2 U}{\partial t^2} = \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2}$

el problema propone aproximar el desplazamiento por medio de: $U_0(x,y,t) = \text{sen}(\pi x)\text{sen}(\pi y)a_0(t)$

Proponemos la misma estructura para realizar la reducción analítica

$$U(x,y,t) = \text{sen}(\pi x)\text{sen}(\pi y)a(t) \quad (3)$$

Calculamos las derivadas respecto a x, y, t

→ $\frac{\partial U}{\partial t} = \text{sen}(\pi x)\text{sen}(\pi y)\,a(t)$ (4) ; $\frac{\partial^2 U}{\partial t^2} = \text{sen}(\pi x)\text{sen}(\pi y)\,a''(t)$

→ $\frac{\partial U}{\partial x} = \pi\cos(\pi x)\text{sen}(\pi y)\,a(t)$; $\frac{\partial^2 U}{\partial x^2} = -\pi^2\text{sen}(\pi x)\text{sen}(\pi y)\,a(t)$

→ $\frac{\partial U}{\partial y} = \pi\,\text{sen}(\pi x)\cos(\pi y)\,a(t)$; $\frac{\partial^2 U}{\partial y^2} = -\pi^2\text{sen}(\pi x)\text{sen}(\pi y)\,a(t)$

como $\frac{\partial^2 U}{\partial t^2} = \frac{\partial^2 U}{\partial x^2} + \frac{\partial^2 U}{\partial y^2}$ necesitamos las segundas derivadas.

$$\text{sen}(\pi x)\text{sen}(\pi y)\,a''(t) = -\pi^2\text{sen}(\pi x)\text{sen}(\pi y)\,a(t) - \pi^2\text{sen}(\pi x)\text{sen}(\pi y)\,a(t)$$

$$\text{sen}(\pi x)\text{sen}(\pi y)\,a''(t) = -2\pi^2\text{sen}(\pi x)\text{sen}(\pi y)\,a(t)$$

$$\text{sen}(\pi x)\text{sen}(\pi y)\,[a''(t) + 2\pi^2 a(t)] = 0$$

$$r(t) = a''(t) + 2\pi^2 a(t) = 0 \quad (4.1)$$

residuo

Incorporamos las c.i

* de la c.i (1): $u(x,y,0) = \text{sen}(\pi x)\text{sen}(\pi y)$

de (3): $u(x,y,0) = \text{sen}(\pi x)\text{sen}(\pi y)\,a(0)$

igualamos y obtenemos $\underline{a(0)=1}$ (5)

* de la c.i (2): $u_t(x,y,0) = \sqrt{2}\pi\,\text{sen}(\pi x)\text{sen}(\pi y)$

y de (4): $u_t(x,y,0) = \text{sen}(\pi x)\,\text{sen}(\pi y)\,a'(0)$

igualamos y obtenemos $\underline{a'(0) = \sqrt{2}\pi}$ (6)

Ahora, debemos resolver $a''(t) + 2\pi^2 a(t) = 0$ (E.D lineal homogénea de 2° orden)

Proponemos una sd: $a(t) = e^{rt}$
$a'(t) = re^{rt}$
$a''(t) = r^2 e^{rt}$

$\Rightarrow r^2 e^{rt} + 2\pi^2 e^{rt} = 0$
$e^{rt}(r^2 + 2\pi^2) = 0$
$r^2 = -2\pi^2$
$r = \pm i\sqrt{2}\pi$ → ¡Frec. angular! $\omega = \sqrt{2}\pi$

La sol. general es $a(t) = C_1\cos(\omega t) + C_2\,\text{sen}(\omega t)$
como calculamos previamente (5) y (6)

- $C_1\cos(0) + C_2\,\text{sen}(0) = 1$
  $C_1(1) + C_2(0) = 1$
  $C_1 = 1$

- $a'(t) = -C_1\omega\,\text{sen}(\omega t) + C_2\omega\cos(\omega t)$
  $\sqrt{2}\pi = -C_1\omega\,\text{sen}(0) + C_2\omega\cos(0)$
  $\sqrt{2}\pi = \omega C_2$
  $\omega = \omega C_2$
  $1 = C_1$

$\Rightarrow a(t) = \cos(\sqrt{2}\pi t) + \text{sen}(\sqrt{2}\pi t)$

y como $u(x,y,t) = \text{sen}(\pi x)\,\text{sen}(\pi y)\,a(t)$
$u(x,y,t) = \text{sen}(\pi x)[\cos(\sqrt{2}\pi t) + \text{sen}(\sqrt{2}\pi t)]$ (7)

Para el Modelo A

$$a_0(t) \sum_{j=1}^{2} A_j \operatorname{sen}(\omega_j t + b_j)$$

$$a_0(t) = A_1 \operatorname{sen}(\omega_1 t + b_1) + A_2 \operatorname{sen}(\omega_2 t + b_2)$$

necesitamos calcular las derivadas del modelo

$$\dot{a_0}(t) = \sum_{j=1}^{2} A_j \omega_j \cos(\omega_j t + b_j)$$

$$\ddot{a_0}(t) = \sum_{j=1}^{2} A_j \omega_j^2 \operatorname{sen}(\omega_j t + b_j)$$

De la ec. (4.1) $r_0(t) = a_0''(t) + 2\pi^2 a_0(t)$

$$r_0(t) = -\sum_{j=1}^{2} A_j \omega_j^2 \operatorname{sen}(\omega_j t + b_j) + 2\pi^2 \sum_{j=1}^{2} A_j \operatorname{sen}(\omega_j t + b_j)$$

$$r_0(t) = \sum_{j=1}^{2} A_j (2\pi^2 - \omega_j^2) \operatorname{sen}(\omega_j t + b_j)$$

$\omega = \sqrt{2\pi} \rightarrow \omega^2 = 2\pi^2$

si la neurona aprende $\rightarrow 2\pi^2 - \omega_j^2 = 0$

y el residuo desaparece

Para el modelo B.

capa 1: $h_1(t) = \operatorname{sen}(\omega_1 t + b_1)$

capa 2: $a_0(t) = A \operatorname{sen}(\omega_2 h_1(t) + b_2)$

sustituimos $a_0(t) = A \operatorname{sen}[\omega_2 \operatorname{sen}(\omega_1 t + b_1) + b_2]$

ahora, calculamos las derivadas

$\rightarrow h_1'(t) = \omega_1 \cos(\omega_1 t + b_1)$

$h_1''(t) = -\omega_1^2 \operatorname{sen}(\omega_1 t + b_1)$

$\rightarrow a_0(t) = A \operatorname{sen}[\omega_2 \operatorname{sen}(\omega_1 t + b_1) + b_2]$

$\dot{a_0}(t) = A\omega_2 \cos[\omega_2 \operatorname{sen}(\omega_1 t + b_1) + b_2] \omega_1 \cos(\omega_1 t + b_1)$

$\ddot{a_0}(t) = -A\omega_1^2 \{\omega_2^2 \operatorname{sen}[\omega_2 \operatorname{sen}(\omega_1 t + b_1) + b_2] \cos^2(\omega_1 t + b_1)$

$+ \omega_2 \cos[\omega_2 \operatorname{sen}(\omega_1 t + b_1) + b_2] \operatorname{sen}(\omega_1 t + b_1)\}$

Para construir el residuo usamos el mismo protocolo que para el modelo B.

$$r_0(t) = a_0''(t) + 2\pi^2 a_0(t)$$

$$r_0(t) = -A\omega_1^2\{\omega_1^2 \operatorname{sen}[\omega_2 \operatorname{sen}(\omega_1 t + b_1) + b_2]\cos^2(\omega_1 t + b_1) + \omega_2 \cos[\omega_2 \operatorname{sen}(\omega_1 t + b_1) + b_2]\operatorname{sen}(\omega_1 t + b_1)\} + 2\pi^2 A \operatorname{sen}[\omega_2 \operatorname{sen}(\omega_1 t + b_1) + b_2]$$

Construimos las funciones de pérdida

Sabemos: $J(\theta) = \frac{1}{m}\sum_{i=1}^{m} r_0(t)^2$

Pero no sería apropiado usar la sumatoria, se necesita que la red satisfaga la ec. desde el inicio hasta el final, por lo tanto, integraremos de $[0, T_F]$ (T_F: tiempo final). Para el error promedio temporal (fun. pérdida)

$$J = \frac{1}{T_F}\int_0^{T_F} r_0(t)^2\, dt$$

④ Red memoria asociativa

Parte analítica

$$E(q) = \frac{1}{2}\sum_{i=1}^{N} q_i^2 - \frac{1}{\beta}\ln\left(\sum_{\mu}^{P} e^{\left[\beta \underbrace{\sum_{i=1}^{N} \gamma_{\mu i} q_i}_{S_\mu}\right]}\right)$$

$$E(q) = \frac{1}{2}\sum_{i=1}^{N} q_i^2 - \frac{1}{\beta}\ln\left(\sum_{\mu}^{P} e^{\beta S_\mu}\right) Z$$

$$E(q) = \frac{1}{2}\sum_{i=1}^{N} q_i^2 - \frac{1}{\beta}\ln Z$$

Calculamos las derivadas parciales con la ec. más simplificada

$$\frac{\partial E}{\partial q_i} = \frac{\partial}{\partial q_i}\left(\frac{1}{2} q_i^2\right) - \frac{1}{\beta}\frac{\partial}{\partial q_i}\ln Z$$

$$= q_i - \frac{1}{\beta}\cdot\frac{1}{Z}\frac{\partial Z}{\partial q_i}$$

Para $\frac{\partial Z}{\partial q_i} = \frac{\partial}{\partial q_i}\left[\sum_{\mu=1}^{P} e^{\beta S_\mu}\right]$

$$= \sum_{\mu=1}^{P} e^{\beta S_\mu} \cdot \beta \frac{\partial S_\mu}{\partial q_i} \rightarrow \sum_{j=1}^{N} \chi_{\mu j} q_j$$

$$= \beta \sum_{\mu=1}^{P} \chi_{\mu j} e^{\beta S_\mu}$$

Retomamos

$$\frac{\partial E}{\partial q_i} = q_i - \frac{1}{\beta} \cdot \frac{1}{Z} \beta \sum_{\mu=1}^{P} \chi_{\mu j} e^{\beta S_\mu}$$

$$\frac{\partial E}{\partial q_i} = q_i - \frac{\sum_{\mu=1}^{P} \chi_{\mu j} e^{\beta \sum \chi_{\mu j} q_j}}{\sum_{\mu=1}^{P} e^{\beta \sum \chi_{\mu j} q_j}}$$

Para deducir la regla de actualización

$$q_i^{(t+1)} = q_i^{(t)} - \alpha \frac{\partial E}{\partial q_i}\Big|_{z = q^{(t)}}$$

$$= q_i^{(t)} - \alpha \left[q_i^{(t)} - \frac{\sum_{\mu}^{P} \chi_{\mu j} e^{\beta \sum \chi_{\mu j} q_j^{(t)}}}{\sum_{\mu}^{P} e^{\beta \sum \chi_{\mu j} q_j^{(t)}}} \right]$$

$$q_i^{(t+1)} = q_i^{(t)} - \alpha q_i^{(t)} + \alpha \frac{\sum_{\mu}^{P} \chi_{\mu j} e^{\sum \chi_{\mu j} q_j^{(t)}}}{\sum_{\mu}^{P} e^{\beta \sum \chi_{\mu j} q_j^{(t)}}}$$

} regla de actualización mediante gradiente descendente.

Si $\alpha = 1$

$$q_i^{(t+1)} = q_i^{(t)} - q_i^{(t)} + \frac{\sum_{\mu}^{P} \chi_{\mu j} e^{\beta \sum \chi_{\mu j} q_j^{(t)}}}{\sum_{\mu}^{P} e^{\beta \sum \chi_{\mu j} q_j^{(t)}}}$$

$$q_i^{(t+1)} = \frac{\sum_{\mu=1}^{P} \chi_{\mu i} e^{\beta \sum \chi_{\mu j} q_j^{(t)}}}{\sum_{\mu=1}^{P} e^{\beta \sum \chi_{\mu j} q_j^{(t)}}} \Rightarrow$$